

Ten Layers of Signed Completion: Growing-Depth Coefficient-Ledger Underdetermination at the Physical First Alias

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Abstract

We audit RH-342–RH-350 and close the batch with a growing-depth information-class theorem. On the physical clock, RH-348 and RH-350 give the exact selected lower-even direct coefficients

$$p_{k,j} = Y_{k,j} + \mathcal{P}_{k,j} - S_{k,j}, \quad m_{k,j} = k - j, \quad 2 \leq j \leq J_k,$$

where $J_k \rightarrow \infty$ and $J_k = o(k)$. RH-350 proves uniform normalized laws for the deterministic demand S and parity packet $\mathcal{P}$, an exact weighted minimax law, and a conditional exponentially divergent subprefix theorem under an unproved aggregate estimate for the actual signed remainder Y .

The new theorem fixes the proved $\mathcal{P}/S$ arrays and considers only their abstract signed coefficient-ledger completions. The affine identity is surjective: every prescribed residual array r is obtained from $Y = S - \mathcal{P} + r$. In particular, $Y^{\text{close}} = S - \mathcal{P}$ gives zero selected residual, while $Y^{\text{far}} = 0$ gives the RH-350 minimax residual. Their normalized budgets obey the exact exchange

$$\mathcal{Y}_k(Y^{\text{close}}) = \mathcal{L}_k(Y^{\text{far}}), \quad \mathcal{Y}_k(Y^{\text{far}}) = \mathcal{L}_k(Y^{\text{close}}) = 0,$$

and $\mathcal{L}_k(Y^{\text{far}})$ has a positive limiting lower bound. Thus the current deterministic/scalar information does not determine the actual selected physical residual. The two completions are not noisy operators, Markov kernels, raw trace partitions, or determinant realizations.

The actual moving-order Y theorem, head transport, critical and first-lower compensation, odd and upper-alias control, and the full E_{off} aggregate remain open. RH-241’s moving noisy all-order envelope remains distinct from the later deterministic anchor and envelope. RH-288 and Gates A–E are inactive. No Riemann-hypothesis conclusion follows.

1 Typed ancestry and the growing window

RH-241 left an all-order deterministic target frontier together with a moving cloud-extracted noisy envelope and coefficient bridge [11]. Later work closes only the deterministic target side. RH-263 proves the deterministic numerator coefficient anchor at every order, RH-267 proves

$$|a_n| < 48q_*^n \quad (n \geq 2), \tag{1}$$

and RH-268 proves $a_n/q_*^n \rightarrow 1$ and the sharp deterministic radius [8, 10, 13]. None of these statements proves the moving noisy uniform envelope, the no-over-extraction bridge, or a moving noisy coefficient theorem. In particular, RH-350’s lower-even triangular uniformity is not the RH-241 moving noisy all-order envelope.

Freeze the physical natural clock and Hardy target

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \eta_k = k - \frac{\log(1/\sigma)}{2 \log \lambda}, \quad H_m = mR^{-2m}, \quad R = \frac{7}{5}. \tag{2}$$

Take an integer triangular window

$$m_{k,j} = k - j, \quad 2 \leq j \leq J_k, \quad J_k \longrightarrow \infty, \quad J_k = o(k). \tag{3}$$

The $o(k)$ restriction is part of the theorem type: it keeps the window in the RH-348 lower-even ladder and permits the RH-350 uniform estimates.

Let $C_M > 0$, $\beta = 1/(r_H \sqrt{\lambda})$, and

$$x = (\beta R)^2 > 1, \quad x\lambda = \left(\frac{R}{r_H}\right)^2 = \left(\frac{28}{17}\right)^2 > 2. \quad (4)$$

With $a_k = C_* C_M \lambda^{\eta_k - 2}$, bounded physical phase makes a_k bounded and positive. RH-350 proves, uniformly on (3),

$$\sup_j \left| \frac{C_M S_{k,j}}{2H_{m_{k,j}} x^{m_{k,j}}} - 1 \right| \rightarrow 0, \quad (5)$$

$$\sup_j \left| \frac{C_M \mathcal{P}_{k,j}}{2H_{m_{k,j}} x^{m_{k,j}}} - a_k \lambda^{2-j} \right| \rightarrow 0. \quad (6)$$

These are deterministic/scalar packet laws. They contain no estimate for the actual signed remainder.

2 Ten source-locked layers

The nine source papers are [1–7, 9, 12]. Their data types and boundaries are retained literally.

paper	strict result	boundary retained here
RH-342	Exact head/counterloop rank lock, shifted-moment recovery under a rank cap, and a hidden $4k$ -shell information-class obstruction.	No actual head-rank identification, root transport, or D_{4k} theorem.
RH-343	Equal rank, squared mass, cap, and fixed-order data admit finite normal spectra with zero or divergent moving prefix budgets.	The models are not noisy operators and do not decide actual head transport.
RH-344	Complete physical critical boundary-orbit extraction and the exact double-alias-sized compensation demand.	The orbit-free rest and head defect are not estimated; no signed critical verdict follows.
RH-345	Conditional physical critical divergence off the scalar balance phase and scalar information-class underdetermination at balance.	The actual $Y_k = o(H_k)$ hypothesis is not proved.
RH-346	Complete physical first-lower boundary-orbit extraction, including the radial sideband and its lower relative scale.	The actual combined remainder remains uncontrolled.
RH-347	Conditional physical first-lower divergence off balance and scalar underdetermination at balance.	The scalar completions are not physical transfer operators and do not decide E_{off} .
RH-348	Exact punctured lower-even orbit ladder, its geometric divergence, and the necessary aggregate signed-supply law.	No theorem estimates the actual signed supply; odd and upper-alias orders remain open.
RH-349	Fixed two-sideband phase incompatibility and exact relative and physically weighted minimax laws.	Both actual remainder hypotheses are unproved; no unconditional prefix statement is made.

paper	strict result	boundary retained here
RH-350	Growing-depth uniform $S/\mathcal{P}$ laws, exact minimax families, and a conditional selected-subprefix obstruction.	The aggregate actual Y hypothesis is unproved; the result is not full E_{off} control.
RH-351	Affine completion surjectivity and opposite growing-depth signed coefficient-ledger completions.	Information-class theorem only; no pair of physical noisy realizations is constructed.

The batch contains ten strict scoped conclusions and zero discharged actual signed-remainder obligations. Exact physical decompositions occur in RH-344, RH-346, and RH-348. The normal-spectral and scalar examples retain their declared information-class type, while the RH-349–RH-350 physical conclusions remain conditional on their named actual remainder hypotheses.

3 The exact selected coefficient and RH-350 input

For each coordinate in (3), RH-348 gives the direct coefficient identity

$$\boxed{p_{k,j} = Y_{k,j} + \mathcal{P}_{k,j} - S_{k,j}}, \quad Y_{k,j} = \mathcal{T}_{k,m_{k,j}}^{\text{rest}} - d_{\sigma,k,2m_{k,j}}. \quad (7)$$

The term Y is signed and actual. It combines the orbit-free physical rest with the defect between the noisy head and the counterloop. It is not a free parameter of the physical problem.

Define the selected direct and remainder budgets

$$\mathcal{L}_k(Y) = \frac{1}{x^{k-2}} \sum_{j=2}^{J_k} \frac{|Y_{k,j} + \mathcal{P}_{k,j} - S_{k,j}|}{2H_{m_{k,j}}}, \quad (8)$$

$$\mathcal{Y}_k(Y) = \frac{1}{x^{k-2}} \sum_{j=2}^{J_k} \frac{|Y_{k,j}|}{2H_{m_{k,j}}}. \quad (9)$$

Since $H_m = mR^{-2m}$, the unnormalized sum in (8) is exactly the selected lower-even contribution with prefix weight $R^{2m}/(2m)$.

Put

$$F_N(a) = \sum_{r=0}^N x^{-r} |a\lambda^{-r} - 1|. \quad (10)$$

RH-350 proves that $a = 1$ is its unique minimizer and

$$A_N := \inf_{a>0} F_N(a) = \frac{1 - x^{-N}}{x - 1} - \frac{1 - (x\lambda)^{-N}}{x\lambda - 1}, \quad (11)$$

$$A_N \nearrow A_\infty = \frac{1}{x - 1} - \frac{1}{x\lambda - 1} > 0. \quad (12)$$

Its conditional physical theorem says

$$\mathcal{Y}_k(Y^{\text{actual}}) \longrightarrow 0 \implies \mathcal{L}_k(Y^{\text{actual}}) = \frac{F_{J_k-2}(a_k)}{C_M} + o(1), \quad (13)$$

and hence $\liminf \mathcal{L}_k(Y^{\text{actual}}) \geq A_\infty/C_M$. The antecedent in (13) is not proved in the repository.

4 Growing-depth signed-completion underdetermination

Definition 4.1 (selected coefficient-ledger information class). Fix the source-locked arrays $S_{k,j}$ and $\mathcal{P}_{k,j}$ satisfying (5)–(6). The selected coefficient-ledger information class consists of formal signed arrays $Y_{k,j}$ on (3), with $p_{k,j}$ defined by (7). No Markov kernel, transfer operator, raw trace partition, determinant, or cross-order physical realizability condition is added.

Proposition 4.2 (affine completion surjectivity). *For every formal residual array $r_{k,j}$ on the selected window, the unique coefficient-ledger completion*

$$Y_{k,j}^{[r]} = S_{k,j} - \mathcal{P}_{k,j} + r_{k,j} \quad (14)$$

satisfies $p_{k,j}^{[r]} = r_{k,j}$ exactly.

Proof. Substitution of (14) into (7) gives $p^{[r]} = (S - \mathcal{P} + r) + \mathcal{P} - S = r$ coordinatewise. Conversely, (7) can be solved uniquely for Y once $p = r$ is prescribed. $\square$

The proposition is deliberately algebraic. It identifies precisely what the proved $S/\mathcal{P}$ information cannot constrain before a theorem for the actual Y is supplied.

Theorem 4.3 (opposite growing-depth coefficient-ledger completions). *Within the selected coefficient-ledger information class, define*

$$Y_{k,j}^{\text{close}} = S_{k,j} - \mathcal{P}_{k,j}, \quad p_{k,j}^{\text{close}} = 0, \quad (15)$$

$$Y_{k,j}^{\text{far}} = 0, \quad p_{k,j}^{\text{far}} = \mathcal{P}_{k,j} - S_{k,j}. \quad (16)$$

Then the normalized budgets obey the exact exchange

$$\boxed{\mathcal{Y}_k(Y^{\text{close}}) = \mathcal{L}_k(Y^{\text{far}}), \quad \mathcal{Y}_k(Y^{\text{far}}) = \mathcal{L}_k(Y^{\text{close}}) = 0.} \quad (17)$$

Moreover, with $N_k = J_k - 2$,

$$\boxed{\mathcal{L}_k(Y^{\text{far}}) = \frac{F_{N_k}(a_k)}{C_M} + o(1),} \quad (18)$$

and

$$\boxed{\liminf_{k \rightarrow \infty} \mathcal{L}_k(Y^{\text{far}}) \geq \frac{1}{C_M} \left(\frac{1}{x-1} - \frac{1}{x\lambda-1} \right) > 0.} \quad (19)$$

Consequently the unnormalized selected sum in the far ledger diverges at least on the exponential scale x^{k-2} , whereas the close ledger vanishes identically on the selected window.

Proof. Equations (15)–(16) are the cases $r = 0$ and $r = \mathcal{P} - S$ of affine surjectivity. The identities $|Y^{\text{close}}| = |S - \mathcal{P}| = |p^{\text{far}}|$ and $Y^{\text{far}} = p^{\text{close}} = 0$ hold coordinatewise. Multiplication by the same positive weights in (8)–(9) and summation prove (17).

For the far ledger, (5)–(6) give, uniformly in j ,

$$\frac{|\mathcal{P}_{k,j} - S_{k,j}|}{2H_{m_{k,j}} x^{m_{k,j}}} = \frac{|a_k \lambda^{2-j} - 1|}{C_M} + o(1).$$

Since $x^{m_{k,j}}/x^{k-2} = x^{2-j}$ and $\sum_{j=2}^{\infty} x^{2-j} < \infty$, summing the uniform error gives $o(1)$. Reindexing by $r = j - 2$ proves (18) directly for the formal ledger. Equations (11)–(12) then give (19). Since $x > 1$, multiplication by x^{k-2} yields the unnormalized divergence. The close statement is exact from (15). $\square$

Corollary 4.4 (the missing Y theorem is decisive). *The proved $S/\mathcal{P}$ arrays, their growing-window uniform laws, and their exact minimax formulas determine neither vanishing nor nonvanishing of the actual selected lower-even residual. Any physical verdict requires an additional moving-order theorem that restricts the actual $Y_{k,j}$.*

Proof. The two formal ledgers satisfy the same fixed $S/\mathcal{P}$ premises and have opposite selected outcomes. Therefore neither outcome is a logical consequence of those premises alone. The close ledger does not contradict RH-350: by (17), its Y budget equals the positive far residual budget and hence fails the small- Y hypothesis. $\square$

Remark 4.5 (strict interpretation). The theorem constructs no physical noisy object. In the physical problem, $Y = \mathcal{T}^{\text{rest}} - d$ is fixed by one noisy dynamics and must satisfy any cross-order identities that such a realization imposes. Those constraints are exactly what the current repository does not estimate. The theorem is a coefficient-information insufficiency result, not a physical existence or nonexistence theorem.

5 Frontier, archive, and claim boundary

The unique repository-backed coordinate after this review is

`actual_growing_lower_even_signed_remainder_open.`

The narrowest positive reopening input is an actual moving-order theorem for $Y_{k,j}$ on (3). Proving $\mathcal{Y}_k(Y^{\text{actual}}) \rightarrow 0$ would activate the RH-350 physical obstruction. Proving weighted closeness of Y^{actual} to $S - \mathcal{P}$ would close only the selected lower-even window. Neither result would by itself estimate the critical order, the first lower sideband, odd orders, upper-alias orders, the head budget D_{4k} , or the complete E_{off} aggregate. A direct physical annular theorem remains an independent possible route.

The executable review records ten scoped conclusions, zero discharged actual signed-remainder obligations, three exact finite algebra witnesses, and twenty-two false forbidden claims. Finite rows reproduce affine identities and budget exchange only. RH-342–RH-350 contain fifteen publication files each and RH-351 contains nineteen, so the batch manifest contains $9 \cdot 15 + 19 = 154$ publication files. After individual and batch archive metadata, the ten controlled paper trees contain 176 files.

The nine upstream result ledgers contain forty-five false Gate values; this review adds five more. All fifty Gate A–E values remain false. RH-288 is inactive because its physical prefix leaf is absent. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

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